EXPERIMENT – 11

AIM:

To develop an Android application using Android Studio that allows users to enter data and store it in the device's SD card (external storage).

ALGORITHM:

1. Input:

The user enters text data in the application.

The user clicks a button to save the data into the SD card.

2. Steps for Implementation in Android Studio:

1. Create a New Android Project:

Open Android Studio

Create a new project with an Empty Activity

Choose Java or Kotlin as the programming language

2. Design the User Interface:

Add the following components in the layout:

o EditText → to enter data

o Button → to save data

o TextView (optional) → to display status messages

3. Add Permissions in Manifest File:

Declare permission to write to external storage

For modern Android versions, also handle runtime permissions

4. Handle Runtime Permissions:

Check if permission is granted

If not, request permission from the user at runtime

Proceed only after permission is granted

5. Check External Storage Availability:

Ensure that:

o SD card is available

o SD card is writable

6. Create File in SD Card:

Define a file name (e.g., mydata.txt)

Select appropriate directory:

o App-specific folder (recommended)

o Or public directory

7. Write Data to SD Card:

Read input from EditText

Open file output stream

Write the data into the file

close the stream

8. Display Output to User:

Show confirmation message:

o “Data saved successfully”

o Or error message if something fails

9. Handle Exceptions:

Manage:

o IOException

o Permission denial

o Storage unavailability

3. Testing the Application:

Run the app on a real device/emulator

Enter sample text

Click Save button

Verify file using file manager in external storage

4. Observations:

Data is stored successfully in the SD card

Runtime permission handling is essential

External storage path must be handled carefully in newer Android versions

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PROGRAM:

Step 1: Create the Project

Step 2:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
```

```
android:background="#ffffff"
android:padding="24dp">

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="SD Card File Operations"
    android:textSize="22sp"
    android:textStyle="bold"
    android:gravity="center"
    android:textColor="#222222"
    android:paddingBottom="8dp"/>

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Write and read data from external storage"
    android:textSize="13sp"
    android:gravity="center"
    android:textColor="#888888"
    android:paddingBottom="24dp"/>

<!-- Input -->
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Enter text to write:"
    android:textSize="14sp"
    android:textColor="#333333"
    android:paddingBottom="6dp"/>
```

```
<EditText
    android:id="@+id/etInput"
    android:layout_width="match_parent"
    android:layout_height="120dp"
    android:hint="Type something here..."
    android:gravity="top"
    android:inputType="textMultiLine"
    android:background="#f5f5f5"
    android:padding="12dp"
    android:layout_marginBottom="16dp"/>

<!-- Write and Read Buttons -->
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:layout_marginBottom="24dp">

    <Button
        android:id="@+id/btnWrite"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="📝 Write to SD"
        android:backgroundTint="#1e88e5"
        android:textColor="#ffffff"
        android:layout_marginEnd="8dp"/>

    <Button
        android:id="@+id/btnRead"
        android:layout_width="0dp"
```

```
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="📖 Read from SD"
        android:backgroundTint="#43a047"
        android:textColor="#ffffff"/>

</LinearLayout>

<!-- Output -->
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="File contents:"
    android:textSize="14sp"
    android:textColor="#333333"
    android:paddingBottom="6dp"/>

<TextView
    android:id="@+id/tvOutput"
    android:layout_width="match_parent"
    android:layout_height="120dp"
    android:text="Nothing read yet..."
    android:textSize="14sp"
    android:textColor="#555555"
    android:background="#f5f5f5"
    android:padding="12dp"
    android:layout_marginBottom="16dp"/>

<!-- File path info -->
<TextView
    android:id="@+id/tvFilePath"
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="File path will appear here"
        android:textSize="11sp"
        android:textColor="#aaaaaa"
        android:gravity="center"/>

    <!-- Clear Button -->
    <Button
        android:id="@+id/btnClear"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Clear"
        android:backgroundTint="#888888"
        android:textColor="#ffffff"
        android:layout_marginTop="12dp"/>

</LinearLayout>
```

Step 3:
Main_Activity.java:

```
package com.example.hostel_management_system;

import android.Manifest;
import android.content.pm.PackageManager;
import android.os.Bundle;
import android.os.Environment;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
```

```
import androidx.core.content.ContextCompat;

import java.io.File;
import java.io.FileInputStream;
import java.io.FileOutputStream;
import java.io.IOException;

public class ThirdActivity extends AppCompatActivity {

    EditText etInput;
    TextView tvOutput, tvFilePath;
    Button btnWrite, btnRead, btnClear;

    private static final int PERMISSION_REQUEST_CODE = 100;
    private static final String FILE_NAME = "hostel_data.txt";

    File file;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_third);

        etInput    = findViewById(R.id.etInput);
        tvOutput   = findViewById(R.id.tvOutput);
        tvFilePath = findViewById(R.id.tvFilePath);
        btnWrite   = findViewById(R.id.btnWrite);
        btnRead    = findViewById(R.id.btnRead);
        btnClear   = findViewById(R.id.btnClear);

        // Set file path — works on API 29+ without permissions
        file = new File(getExternalFilesDir(null), FILE_NAME);
        tvFilePath.setText("Path: " + file.getAbsolutePath());

        // Request permissions for older Android versions
        checkPermissions();
```

```
    btnWrite.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            String text = etInput.getText().toString().trim();
            if (text.isEmpty()) {
                Toast.makeText(ThirdActivity.this,
                    "Please enter some text first", Toast.LENGTH_SHORT).show();
                return;
            }
            writeToFile(text);
        }
    });

    btnRead.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            readFromFile();
        }
    });

    btnClear.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            etInput.setText("");
            tvOutput.setText("Nothing read yet...");
            Toast.makeText(ThirdActivity.this,
                "Cleared", Toast.LENGTH_SHORT).show();
        }
    });
}

private void writeToFile(String text) {
    try {
        FileOutputStream fos = new FileOutputStream(file);
        fos.write(text.getBytes());
```

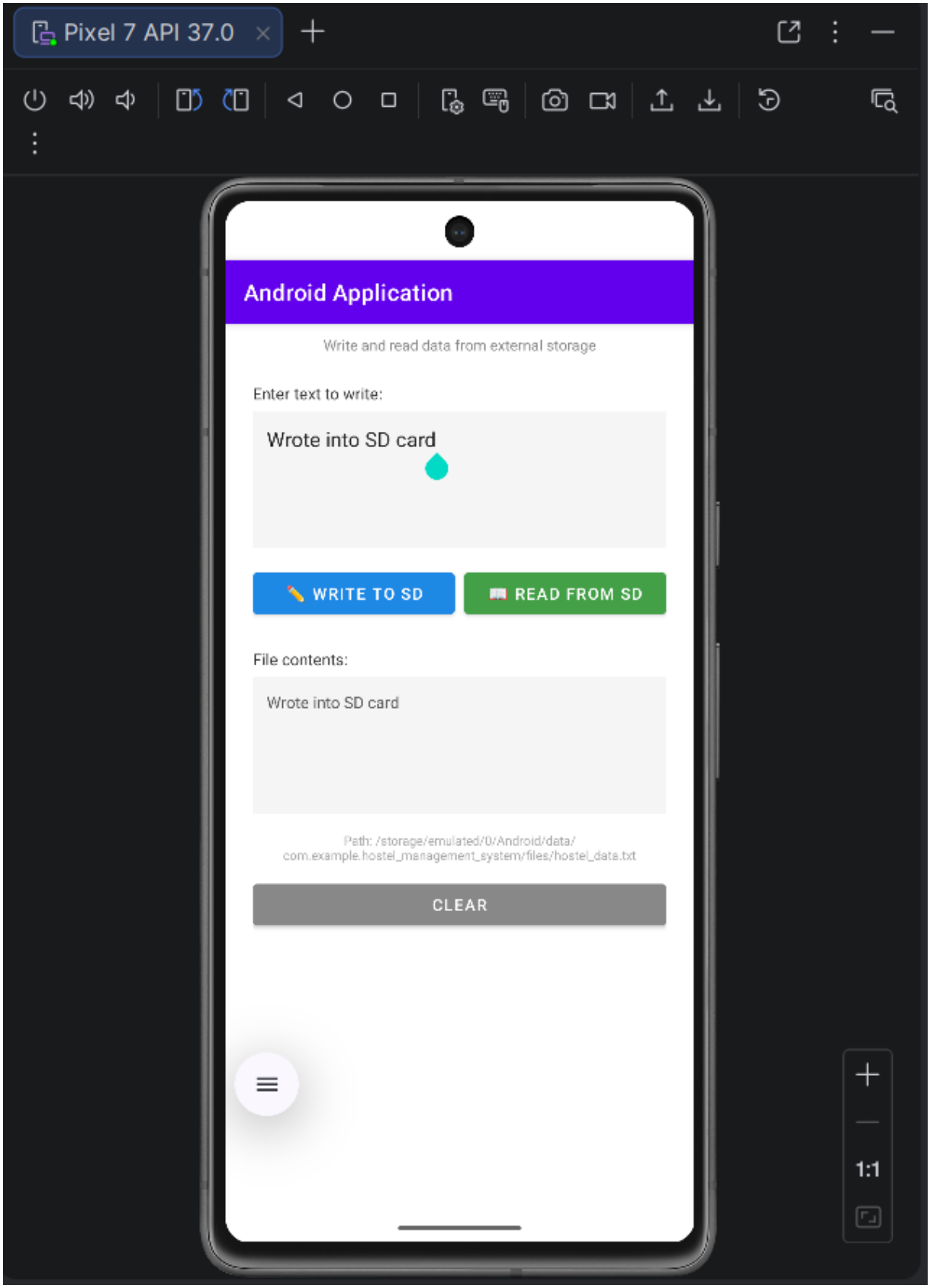
```
        fos.close();
        Toast.makeText(this,
            "Data written to SD card successfully!", Toast.LENGTH_SHORT).show();
    } catch (IOException e) {
        Toast.makeText(this,
            "Write failed: " + e.getMessage(), Toast.LENGTH_LONG).show();
    }
}

private void readFromFile() {
    if (!file.exists()) {
        Toast.makeText(this,
            "No file found. Write something first.", Toast.LENGTH_SHORT).show();
        return;
    }
    try {
        FileInputStream fis = new FileInputStream(file);
        byte[] buffer = new byte[(int) file.length()];
        fis.read(buffer);
        fis.close();
        String content = new String(buffer);
        tvOutput.setText(content);
        Toast.makeText(this,
            "Data read successfully!", Toast.LENGTH_SHORT).show();
    } catch (IOException e) {
        Toast.makeText(this,
            "Read failed: " + e.getMessage(), Toast.LENGTH_LONG).show();
    }
}

private void checkPermissions() {
    if (ContextCompat.checkSelfPermission(this,
            Manifest.permission.WRITE_EXTERNAL_STORAGE)
            != PackageManager.PERMISSION_GRANTED) {
        ActivityCompat.requestPermissions(this,
            new String[]{
```

```
            Manifest.permission.WRITE_EXTERNAL_STORAGE,
            Manifest.permission.READ_EXTERNAL_STORAGE
        },
        PERMISSION_REQUEST_CODE);
    }
  }
}
```

OUTPUT:



Result: An Android application is successfully developed using Android Studio to write data to the SD card with proper permission handling and storage management.